



Contrastive Learning Through Time

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')

import os

BASE = '/content/drive/MyDrive/CLTT_Soccer'
FRAMES_DRIVE = f'{BASE}/frames'
CKPT_DIR = f'{BASE}/checkpoints'
UCF_DIR = f'{BASE}/ucf101'

for d in [BASE, FRAMES_DRIVE, CKPT_DIR, UCF_DIR]:
    os.makedirs(d, exist_ok=True)

print("Drive mounted. Directories created.")
```

Mounted at /content/drive
Drive mounted. Directories created.

```
In [2]: !pip install -q yt-dlp decord av
```

```
0 182.3/182.3 kB 14.7 MB/s eta 0:00:00
3.3/3.3 MB 128.2 MB/s eta 0:00:00
13.6/13.6 MB 144.7 MB/s eta 0:00:00
36.3/36.3 MB 77.2 MB/s eta 0:00:00
```

```
In [ ]: VIDEO_URL = "https://www.youtube.com/watch?v=gTmR5PMIIjs"
VIDEO_PATH = f'{BASE}/soccer_asmr.mp4'

if not os.path.exists(VIDEO_PATH):
    !yt-dlp -f "bestvideo[height<=720][ext=mp4]+bestaudio[ext=m4a]/best[height
else:
    print("Video already on Drive, skipping download.")
```

Video already on Drive, skipping download.

```
In [4]: LOCAL_VIDEO = '/content/soccer_asmr.mp4'

if not os.path.exists(LOCAL_VIDEO):
    print("Copying video to local SSD...")
    !cp "{VIDEO_PATH}" "{LOCAL_VIDEO}"
    print(f"Done: {os.path.getsize(LOCAL_VIDEO)/1e9:.2f} GB")

print("Video ready.")
```

Copying video to local SSD...
Done: 2.15 GB
Video ready.

Frame Extraction

Frames are extracted at 5 fps and scaled to 128×128. 5 fps avoids storing near-duplicate frames while still giving each 8-frame clip (~1.6s) enough motion information. The 128×128 resolution is cropped to 112×112 during augmentation.

```
In [ ]: import os

os.makedirs('/content/soccer_frames', exist_ok=True)

# Video is already local – ffmpeg reads it sequentially (very fast)
!ffmpeg -i /content/soccer_asmr.mp4 -vf "fps=5,scale=128:128" -q:v 3 /content/

n = len([f for f in os.listdir('/content/soccer_frames') if f.endswith('.jpg')])
print(f"Extracted {n} frames")
```

Model Architecture and Loss

CLTTDataset samples positive clip pairs: `clip_a` at frame `idx`, `clip_b` at `idx + offset` where `offset ~ Uniform(min_offset, pos_window)`. The 2s positive window bounds how far apart the clips can be; the 0.8s minimum offset prevents full overlap (at 5 fps: window = 10 frames, min_offset = 4 frames).

CLTTModel uses R(2+1)D-18, which factorizes 3D convolutions into separate spatial (2D) and temporal (1D) passes — more parameter-efficient than full 3D convolutions while still capturing spatiotemporal features. The two-layer MLP projection head maps the 512-dim encoder output to 128 dims for the contrastive loss; following SimCLR, this head is discarded at evaluation time.

NTXentLoss computes NT-Xent over a $2B \times 2B$ similarity matrix. Temperature $\tau = 0.07$ sharpens the softmax, pushing the model to strongly separate non-paired clips rather than settling for a diffuse embedding space.

```
In [ ]: import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
import torchvision.transforms as T
import torchvision.models.video as video_models
from PIL import Image
import random, numpy as np
from tqdm import tqdm

# Kinetics-400 normalization stats
MEAN = [0.43216, 0.39467, 0.37645]
STD = [0.22803, 0.22145, 0.21699]
```

```

from PIL import Image
import os, random, torch
from torch.utils.data import Dataset

class CLTTDataset(Dataset):
    def __init__(self, frames_dir, clip_len=8, fps=5,
                 pos_window_sec=2, min_offset_sec=0.8, transform=None):
        self.frames_dir = frames_dir
        self.clip_len = clip_len
        self.pos_window = int(pos_window_sec * fps)  # max offset in frames
        self.min_offset = max(1, int(min_offset_sec * fps))  # min offset in
        self.transform = transform
        self.frames = sorted(f for f in os.listdir(frames_dir)
                             if f.endswith('.jpg'))
        self.N = len(self.frames)
        print(f"{self.N} frames | {self.N/fps/3600:.2f} hrs | "
              f"pos_window={pos_window_sec}s | min_offset={min_offset_sec}s")

    def _load_clip(self, start):
        imgs = []
        for i in range(self.clip_len):
            idx = min(start + i, self.N - 1)
            img = Image.open(f'{self.frames_dir}/{self.frames[idx]}').convert('RGB')
            imgs.append(self.transform(img) if self.transform else img)
        return torch.stack(imgs, dim=1)  # C, T, H, W

    def __len__(self):
        return max(0, self.N - self.clip_len - self.pos_window - 1)

    def __getitem__(self, idx):
        clip_a = self._load_clip(idx)
        offset = random.randint(self.min_offset, self.pos_window)
        clip_b = self._load_clip(min(idx + offset, self.N - self.clip_len - 1))
        return clip_a, clip_b

class CLTTModel(nn.Module):
    """R(2+1)D-18 encoder + MLP projection head."""
    def __init__(self, proj_dim=128):
        super().__init__()
        backbone = video_models.r2plus1d_18(weights=None)

        self.stem = backbone.stem
        self.layer1 = backbone.layer1
        self.layer2 = backbone.layer2
        self.layer3 = backbone.layer3
        self.layer4 = backbone.layer4
        self.avgpool = backbone.avgpool  # → (B, 512, 1, 1, 1)

        self.projector = nn.Sequential(

```

```

        nn.Linear(512, 512),
        nn.BatchNorm1d(512),
        nn.ReLU(inplace=True),
        nn.Linear(512, proj_dim),
    )

    def encode(self, x):
        x = self.stem(x)
        x = self.layer1(x)
        x = self.layer2(x)
        x = self.layer3(x)
        x = self.layer4(x)
        x = self.avgpool(x)
        return x.flatten(1)  # (B, 512)

    def forward(self, x):
        z = self.projector(self.encode(x))
        return F.normalize(z, dim=-1)  # (B, proj_dim)

class NTXentLoss(nn.Module):
    """NT-Xent (Normalized Temperature-scaled Cross Entropy)."""
    def __init__(self, temperature=0.07):
        super().__init__()
        self.T = temperature

    def forward(self, z_a, z_b):
        B = z_a.size(0)
        z = torch.cat([z_a, z_b], dim=0)  # (2B, D)
        sim = torch.mm(z, z.T) / self.T  # (2B, 2B)

        mask = torch.eye(2 * B, dtype=torch.bool, device=z.device)
        sim = sim.masked_fill(mask, float('-inf'))

        labels = torch.cat([
            torch.arange(B, 2 * B, device=z.device),
            torch.arange(0, B, device=z.device),
        ])
        return F.cross_entropy(sim, labels)

print("Classes defined.")

```

Pre-training

Augmentations (RandomCrop, RandomHorizontalFlip, ColorJitter) are applied independently to each clip in a pair. This forces the encoder to match semantically similar clips despite different appearance — preventing it from exploiting low-level pixel statistics. `drop_last=True` avoids a final batch of size 1 that would break BatchNorm. AdamW with CosineAnnealingLR decays the learning rate smoothly across all batches in both epochs.

```

In [ ]: import torch.optim as optim
        from torch.cuda.amp import autocast, GradScaler
        import json

def run_pretraining(frames_dir, ckpt_dir, epochs=2, batch_size=32, lr=3e-4):
    device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
    print(f"Device: {device}")

    train_transform = T.Compose([
        T.RandomCrop(112),
        T.RandomHorizontalFlip(),
        T.ColorJitter(brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1),
        T.ToTensor(),
        T.Normalize(MEAN, STD),
    ])

    dataset = CLTTDataset(frames_dir, clip_len=8, fps=5,
                          pos_window_sec=2, min_offset_sec=0.8,
                          transform=train_transform)

    loader = DataLoader(dataset, batch_size=32, shuffle=True,
                       num_workers=4, pin_memory=True, drop_last=True)

    model = CLTTModel(proj_dim=128).to(device)
    criterion = NTXentLoss(temperature=0.07)
    optimizer = optim.AdamW(model.parameters(), lr=lr, weight_decay=1e-4)
    scheduler = optim.lr_scheduler.CosineAnnealingLR(
        optimizer, T_max=epochs * len(loader))
    scaler = torch.cuda.amp.GradScaler('cuda')

    loss_history = {'batch': [], 'epoch': []}

    for epoch in range(epochs):
        model.train()
        running_loss = 0.0
        pbar = tqdm(loader, desc=f'Epoch {epoch+1}/{epochs}')

        for clip_a, clip_b in pbar:
            clip_a, clip_b = clip_a.to(device), clip_b.to(device)

            optimizer.zero_grad()
            with autocast():
                z_a = model(clip_a)
                z_b = model(clip_b)
                loss = criterion(z_a, z_b)

            scaler.scale(loss).backward()
            scaler.step(optimizer)
            scaler.update()
            scheduler.step()

            batch_loss = loss.item()
            running_loss += batch_loss

```

```

        loss_history['batch'].append(batch_loss)
        pbar.set_postfix(loss=f'{batch_loss:.4f}')

    avg = running_loss / len(loader)
    loss_history['epoch'].append(avg)
    print(f'Epoch {epoch+1} - avg loss: {avg:.4f}')

    ckpt = f'{ckpt_dir}/cltt_epoch{epoch+1}.pt'
    torch.save({'epoch': epoch+1,
                'model_state_dict': model.state_dict(),
                'loss': avg}, ckpt)
    print(f'Saved → {ckpt}')

    with open(f'{ckpt_dir}/loss_history.json', 'w') as f:
        json.dump(loss_history, f)
    print("Loss history saved.")

    return model

print("Pretraining function ready.")

```

In [30]: LOCAL_FRAMES = '/content/soccer_frames'

```

pretrained_model = run_pretraining(
    frames_dir = LOCAL_FRAMES,
    ckpt_dir    = CKPT_DIR,
    epochs      = 2,
    batch_size  = 32,
    lr          = 3e-4,
)

```

Device: cuda

55075 frames | 3.06 hrs | pos_window=2s | min_offset=0.8s

Epoch 1/2: 0%| | 0/1720 [00:00<?, ?it/s]/tmp/ipykernel_796/69783381.py:42: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.

with autocast():

/tmp/ipykernel_796/69783381.py:50: UserWarning: Detected call of `lr\_scheduler.step()` before `optimizer.step()`. In PyTorch 1.1.0 and later, you should call them in the opposite order: `optimizer.step()` before `lr\_scheduler.step()`. Failure to do this will result in PyTorch skipping the first value of the learning rate schedule. See more details at <https://pytorch.org/docs/stable/optim.html#how-to-adjust-learning-rate>

scheduler.step()

Epoch 1/2: 100%|██████████| 1720/1720 [09:37<00:00, 2.98it/s, loss=0.4965]

Epoch 1 - avg loss: 0.6132

Saved → /content/drive/MyDrive/CLTT_Soccer/checkpoints/cltt_epoch1.pt

Epoch 2/2: 100%|██████████| 1720/1720 [09:41<00:00, 2.96it/s, loss=0.0591]

Epoch 2 - avg loss: 0.2293

Saved → /content/drive/MyDrive/CLTT_Soccer/checkpoints/cltt_epoch2.pt

Loss history saved.

In [31]: import json, matplotlib.pyplot as plt

```

import numpy as np

with open(f'{CKPT_DIR}/loss_history.json') as f:
    history = json.load(f)

batch_losses = history['batch']
epoch_losses = history['epoch']
n_epochs = len(epoch_losses)
epoch_size = len(batch_losses) // n_epochs

# Smooth batch losses with a rolling average for readability
window = max(1, epoch_size // 20)
smooth = np.convolve(batch_losses, np.ones(window) / window, mode='valid')
x_smooth = np.linspace(1, n_epochs, len(smooth))

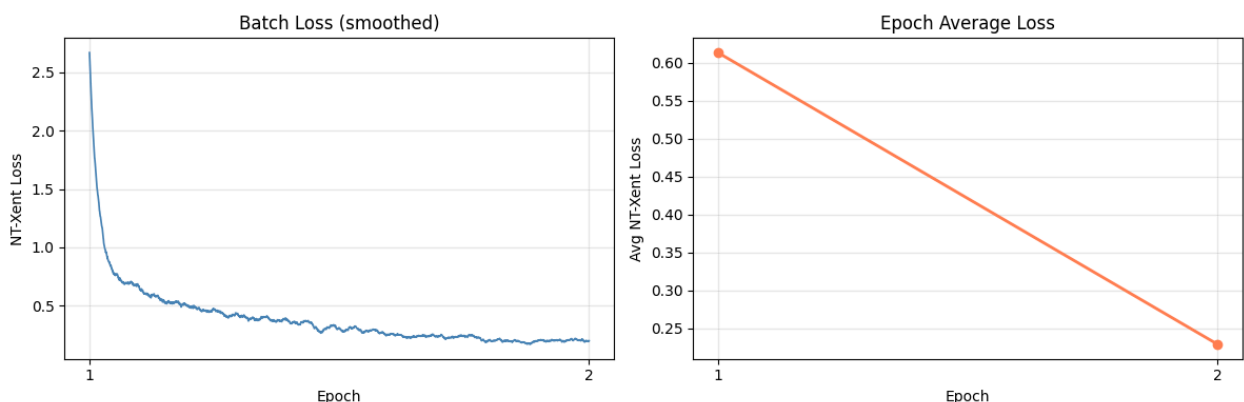
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Left: smoothed batch loss
axes[0].plot(x_smooth, smooth, linewidth=1.2, color='steelblue')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('NT-Xent Loss')
axes[0].set_title('Batch Loss (smoothed)')
axes[0].set_xticks(range(1, n_epochs + 1))
axes[0].grid(alpha=0.3)

# Right: per-epoch average
axes[1].plot(range(1, n_epochs + 1), epoch_losses,
             marker='o', linewidth=2, color='coral')
axes[1].set_xlabel('Epoch')
axes[1].set_ylabel('Avg NT-Xent Loss')
axes[1].set_title('Epoch Average Loss')
axes[1].set_xticks(range(1, n_epochs + 1))
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.savefig(f'{CKPT_DIR}/loss_curve.png', dpi=150)
plt.show()
print("Plot saved to Drive.")

```



Plot saved to Drive.

In [32]: `import shutil`

```

ucf101_dir = f'{UCF_DIR}/UCF-101'

if os.path.exists(ucf101_dir):
    n = sum(len(os.listdir(f'{ucf101_dir}/{c}')) for c in os.listdir(ucf101_dir))
    print(f"UCF-101 ready – {n} videos across {len(os.listdir(ucf101_dir))} classes.")
else:
    split_dirs_exist = any(os.path.exists(f'{UCF_DIR}/{s}') for s in ['train', 'test', 'val'])
    if not split_dirs_exist:
        zip_path = f'{UCF_DIR}/archive.zip'
        if not os.path.exists(zip_path):
            raise FileNotFoundError(f"archive.zip not found at {zip_path} – up")
        print("Extracting UCF101 from Drive archive.zip...")
        !unzip -oq "{zip_path}" -d "{UCF_DIR}/"

    print("Merging train/test/val into UCF-101/ (one-time, may take a few minutes)")
    os.makedirs(ucf101_dir, exist_ok=True)
    for split in ['train', 'test', 'val']:
        split_dir = f'{UCF_DIR}/{split}'
        if not os.path.exists(split_dir):
            continue
        for cls in os.listdir(split_dir):
            cls_dst = f'{ucf101_dir}/{cls}'
            os.makedirs(cls_dst, exist_ok=True)
            for video in os.listdir(f'{split_dir}/{cls}'):
                src = f'{split_dir}/{cls}/{video}'
                dst = f'{cls_dst}/{video}'
                if not os.path.exists(dst):
                    shutil.move(src, dst)

    n = sum(len(os.listdir(f'{ucf101_dir}/{c}')) for c in os.listdir(ucf101_dir))
    print(f"Done – {n} videos across {len(os.listdir(ucf101_dir))} classes.")

```

UCF-101 ready – 13451 videos across 101 classes.

```

In [33]: splits_dir = f'{UCF_DIR}/ucfTrainTestlist'
n_splits = len(os.listdir(splits_dir)) if os.path.exists(splits_dir) else 0

if n_splits < 6:
    drive_zip = f'{UCF_DIR}/splits.zip'
    if os.path.exists(drive_zip):
        print("Found splits.zip on Drive – extracting...")
        !unzip -o "{drive_zip}" -d /tmp/ucf_splits_dir/
    else:
        print("Downloading UCF101 train/test splits from UCF website...")
        !wget -c "https://www.crcv.ucf.edu/data/UCF101/UCF101TrainTestSplits-Release101_v2015.zip" -O /tmp/ucf_splits.zip
        !unzip -o /tmp/ucf_splits.zip -d /tmp/ucf_splits_dir/
        !cp -rf /tmp/ucf_splits_dir/ucfTrainTestlist "{UCF_DIR}/"
        print(f"Splits ready ({len(os.listdir(splits_dir))} files).")
    else:
        print(f"Splits already present ({n_splits} files).")

```

Splits already present (7 files).

Downstream Evaluation: Feature Extraction on UCF-101

The pre-trained encoder is run over all UCF-101 train and test clips with weights frozen, producing fixed 512-dim feature vectors. Keeping the encoder frozen isolates representation quality from any supervised fine-tuning signal.

```
In [34]: import glob
from torchvision.datasets import UCF101

device = torch.device('cuda')

# Load pretrained model from checkpoint if not already in memory
if 'pretrained_model' not in vars() or pretrained_model is None:
    ckpts = sorted(glob.glob(f'{CKPT_DIR}/cltt_epoch*.pt'))
    if not ckpts:
        raise FileNotFoundError(f"No checkpoints in {CKPT_DIR} – run pretraini
    latest = ckpts[-1]
    print(f"Loading checkpoint: {latest}")
    pretrained_model = CLTTModel(proj_dim=128).to(device)
    state = torch.load(latest, map_location=device)
    pretrained_model.load_state_dict(state['model_state_dict'])
    pretrained_model.eval()
    print(f"Loaded epoch {state['epoch']}, loss {state['loss']:.4f}")

def make_ucf_transform():
    return T.Compose([
        T.Resize(128),
        T.CenterCrop(112),
        T.ConvertImageDtype(torch.float32),
        T.Normalize(MEAN, STD),
    ])

def ucf_collate(batch):
    videos = torch.stack([item[0] for item in batch])
    labels = torch.tensor([item[2] for item in batch])
    return videos, None, labels

def load_ucf101(ucf_dir, split='train', fold=1):
    return UCF101(
        root = f'{ucf_dir}/UCF-101',
        annotation_path = f'{ucf_dir}/ucfTrainTestlist',
        frames_per_clip = 16,
        step_between_clips = 8,
        fold = fold,
        train = (split == 'train'),
        transform = make_ucf_transform(),
        output_format = 'TCHW',
    )

@torch.no_grad()
```

```

def extract_features(model, loader, device):
    model.eval()
    features, labels = [], []
    for videos, _, lbls in tqdm(loader, desc='Extracting'):
        x = videos.permute(0, 2, 1, 3, 4).to(device)
        h = model.encode(x).cpu()
        features.append(h)
        labels.append(lbls)
    return torch.cat(features), torch.cat(labels)

train_ds = load_ucf101(UCF_DIR, 'train')
test_ds = load_ucf101(UCF_DIR, 'test')

train_loader = DataLoader(train_ds, batch_size=16, num_workers=4, collate_fn=collate_fn)
test_loader = DataLoader(test_ds, batch_size=16, num_workers=4, collate_fn=collate_fn)

print("Extracting train features...")
train_feats, train_labels = extract_features(pretrained_model, train_loader, device)
print("Extracting test features...")
test_feats, test_labels = extract_features(pretrained_model, test_loader, device)

print(f"Train: {train_feats.shape}, Test: {test_feats.shape}")
torch.save({'feats': train_feats, 'labels': train_labels}, f'{CKPT_DIR}/train_feats_labels.pt')
torch.save({'feats': test_feats, 'labels': test_labels}, f'{CKPT_DIR}/test_feats_labels.pt')
print("Features saved.")

```

```

0%|          | 0/841 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/
torchvision/io/_video_deprecation_warning.py:9: UserWarning: The video decoding
and encoding capabilities of torchvision are deprecated from version 0.22 and w
ill be removed in version 0.24. We recommend that you migrate to TorchCodec, wh
ere we'll consolidate the future decoding/encoding capabilities of PyTorch: htt
ps://github.com/pytorch/torchcodec
  warnings.warn(
100%|██████████| 841/841 [06:22<00:00, 2.20it/s]
0%|          | 0/841 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/
torchvision/io/_video_deprecation_warning.py:9: UserWarning: The video decoding
and encoding capabilities of torchvision are deprecated from version 0.22 and w
ill be removed in version 0.24. We recommend that you migrate to TorchCodec, wh
ere we'll consolidate the future decoding/encoding capabilities of PyTorch: htt
ps://github.com/pytorch/torchcodec
  warnings.warn(
100%|██████████| 841/841 [06:19<00:00, 2.21it/s]
Extracting train features...

```

```
Extracting: 0%|          | 0/13099 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warning.py:9: UserWarning: The video decoding and encoding capabilities of torchvision are deprecated from version 0.22 and will be removed in version 0.24. We recommend that you migrate to TorchCodec, where we'll consolidate the future decoding/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
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  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarning: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarning: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarning: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarning: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.")
Extracting: 100%|██████████| 13099/13099 [1:39:31<00:00, 2.19it/s]
Extracting test features...
```

```

Extracting: 0%|          | 0/5109 [00:00<?, ?it/s]/usr/local/lib/python3.12/d
ist-packages/torchvision/io/_video_deprecation_warning.py:9: UserWarning: The v
ideo decoding and encoding capabilities of torchvision are deprecated from vers
ion 0.22 and will be removed in version 0.24. We recommend that you migrate to
TorchCodec, where we'll consolidate the future decoding/encoding capabilities o
f PyTorch: https://github.com/pytorch/torchcodec
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ng.py:9: UserWarning: The video decoding and encoding capabilities of torchvisi
on are deprecated from version 0.22 and will be removed in version 0.24. We rec
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  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warni
ng.py:9: UserWarning: The video decoding and encoding capabilities of torchvisi
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ing/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warni
ng.py:9: UserWarning: The video decoding and encoding capabilities of torchvisi
on are deprecated from version 0.22 and will be removed in version 0.24. We rec
ommend that you migrate to TorchCodec, where we'll consolidate the future decod
ing/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
Extracting: 100%|██████████| 5109/5109 [39:41<00:00, 2.14it/s]
Train: torch.Size([209582, 512]), Test: torch.Size([81743, 512])
Features saved.

```

Linear Probe

A single `Linear(512, 101)` layer is trained on the frozen encoder features. No encoder weights are updated. Linear probe accuracy is the standard measure for self-supervised representation quality — if the features are linearly separable, the encoder has learned meaningful structure without supervision.

```

In [ ]: from torch.utils.data import TensorDataset

# Load from Drive if needed
train_data = torch.load(f'{CKPT_DIR}/train_feats.pt')
test_data = torch.load(f'{CKPT_DIR}/test_feats.pt')
train_feats, train_labels = train_data['feats'], train_data['labels']
test_feats, test_labels = test_data['feats'], test_data['labels']

NUM_CLASSES = 101

# Linear probe: frozen features -> trainable linear head
probe = nn.Linear(512, NUM_CLASSES).to(device)
opt = optim.AdamW(probe.parameters(), lr=1e-3, weight_decay=1e-4)

probe_train = DataLoader(TensorDataset(train_feats, train_labels),
                          batch_size=256, shuffle=True)
probe_test = DataLoader(TensorDataset(test_feats, test_labels),
                        batch_size=256)

for epoch in range(20):
    probe.train()
    for x, y in probe_train:
        x, y = x.to(device), y.to(device)
        loss = F.cross_entropy(probe(x), y)
        opt.zero_grad()
        loss.backward()
        opt.step()

    # Eval every 5 epochs
    if (epoch + 1) % 5 == 0:
        probe.eval()
        correct = total = 0
        with torch.no_grad():
            for x, y in probe_test:
                x, y = x.to(device), y.to(device)
                preds = probe(x).argmax(dim=1)
                correct += (preds == y).sum().item()
                total += y.size(0)
        print(f"Epoch {epoch+1:2d} - Test Acc: {100*correct/total:.2f}%")

```

```

Epoch 5 - Test Acc: 22.17%
Epoch 10 - Test Acc: 22.09%
Epoch 15 - Test Acc: 22.39%
Epoch 20 - Test Acc: 23.04%

```